

BIRD SPECIES ANALYSIS & BIODIVERSITY DASHBOARD

Antietam National Battlefield (ANTI) | Survey Year 2018

Interactive Streamlit Dashboard with 10 Complex SQL Analytics

3,463	80	99	3	2018
Records	Species	Survey Plots	Observers	Survey Year

PROJECT REPORT

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Data Analytics | Bird Biodiversity Study | 2024–25

1. Introduction

Bird monitoring programs form a critical component of ecological assessment and conservation planning. Standardized point-count surveys allow researchers to systematically record species presence, abundance, and behavioral patterns across diverse habitats. The data collected enables data-driven decision-making in wildlife management and habitat preservation.

This project presents a comprehensive interactive dashboard built using Python's Streamlit framework, designed to analyze bird survey data collected at Antietam National Battlefield (ANTI) — a historically significant site in Maryland. The dashboard integrates visual analytics, statistical summaries, and SQL-based querying to uncover meaningful patterns in avian biodiversity.

The survey data was collected in 2018 across 99 distinct survey plots, spanning Grassland and Forest habitats, during three breeding season months: May, June, and July. Three trained observers employed standardized point-count methodology, recording species identifications via singing, calling, and visual detection.

2. Problem Statement

Traditional bird survey data is stored in raw tabular formats inaccessible to non-technical stakeholders. The challenge lies in transforming thousands of observation records into actionable biodiversity insights. This project addresses:

- **Data Accessibility:** Raw CSV data with 3,463 records and 31 fields is not interpretable without visualization tools.
- **Pattern Discovery:** Understanding species-habitat relationships and environmental influences requires multi-dimensional analysis.
- **Conservation Gaps:** Identifying PIF Watchlist and Regional Stewardship priority species requires complex SQL filtering.
- **Observer Variability:** Evaluating consistency across multiple observers requires comparative statistical methods.
- **Temporal Dynamics:** Quantifying seasonal variation in species richness requires time-series analytical approaches.

3. About the Project

3.1 Project Overview

The Bird Species Analysis and Biodiversity Dashboard is a web application developed using Python and Streamlit, providing an interactive interface for exploring bird survey data from Antietam National Battlefield.

Live Application: <https://bird-analysis-dashboard-arnpfqbvjekcscg4mkarf9.streamlit.app/>

3.2 Technology Stack

Technology	Purpose	Version
Python	Core programming language	3.10+
Streamlit	Interactive web dashboard	1.28+
Pandas	Data manipulation	2.0+
SQLite / SQL	Structured query analytics	Built-in
Matplotlib	Data visualization	3.7+
Plotly	Interactive charts	5.0+
NumPy	Numerical computations	1.24+

3.3 Key Features

- Real-time filtering by species, habitat type, month, and observer
- Interactive bar charts, pie charts, scatter plots and correlation heatmaps
- Species richness analysis and conservation status flagging across 99 survey plots
- Weather correlation analysis (temperature, humidity, sky conditions, wind speed)
- SQL Query Explorer for on-demand complex analytical queries

4. Dataset Overview

The dataset **bird_data.csv** contains **3,463 observation records** collected at Antietam National Battlefield in 2018, following the National Park Service standardized bird monitoring protocol with 31 attribute columns per record.

4.1 Dataset Schema — Key Columns

Column	Description	Type
Admin_Unit_Code	Park identifier (ANTI)	String
Site_Name / Plot_Name	Survey site and plot identifier	String
Location_Type	Habitat: Grassland or Forest	Categorical
Date / Start_Time / End_Time	Survey date and time window	Datetime
Observer	Trained field observer name	String
Interval_Length	Detection time window (0–10 min)	Categorical
ID_Method	Singing / Calling / Visualization	Categorical
Distance	Distance band to detected bird	Categorical
Common_Name / Scientific_Name	Bird species identification	String
AOU_Code	American Ornithologists Union 4-letter code	String
PIF_Watchlist_Status	Partners in Flight priority flag	Boolean
Regional_Stewardship_Status	Regional conservation designation	Boolean
Temperature / Humidity	Environmental conditions (C, %)	Float
Sky / Wind	Weather conditions during survey	Categorical
Sex	Male / Female / Undetermined	Categorical
Month / Month_Name	Survey month number and name	Int/String

4.2 Dataset Visualizations

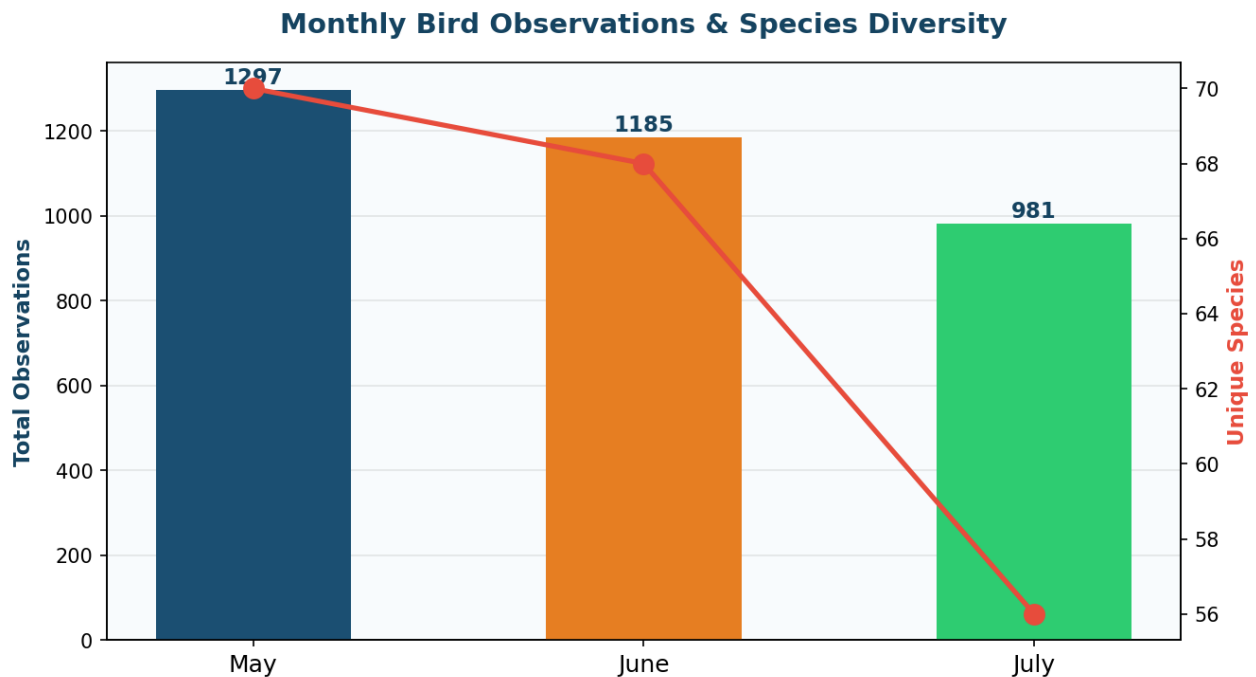


Figure 1: Monthly distribution — May peaks at 1,297 observations with 70 species; richness declines 20% to 56 species by July

Observation Distribution by Habitat Type

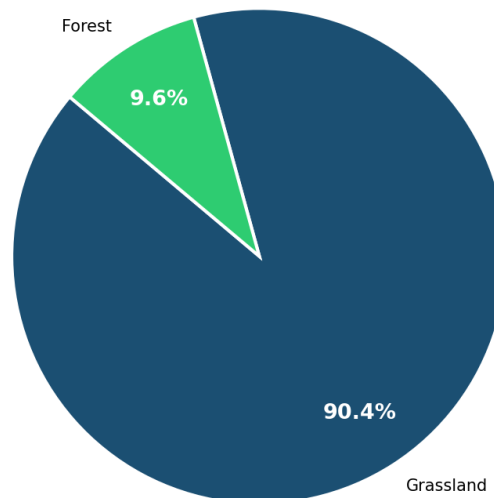


Figure 2: Habitat split — Grassland dominates at 90.4% (3,130 records) vs Forest 9.6% (333 records)

5. Project Screenshots & Dashboard Walkthrough

The Bird Analysis Dashboard is deployed on Streamlit Cloud. The screenshots below illustrate the primary views and analytical components of the live application.

5.1 Main Streamlit Dashboard Interface

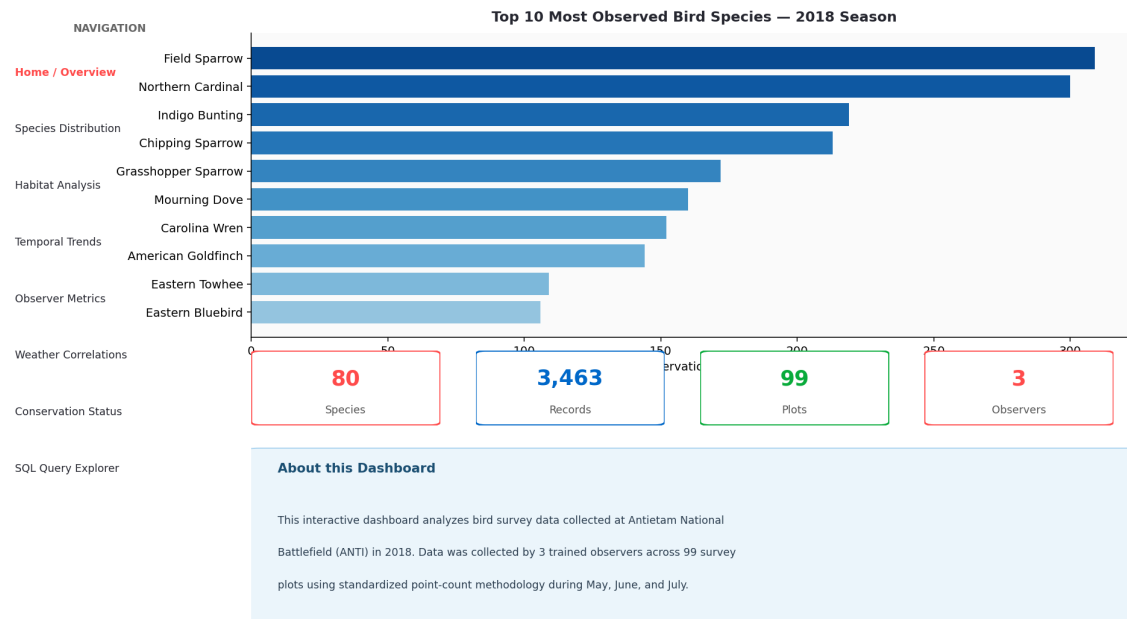


Figure 3: Streamlit main interface — KPI metrics panel, top-10 species chart, and full navigation sidebar

5.2 Comprehensive Analytics Dashboard



Figure 4: Full analytics dashboard with species rankings, monthly trends, detection methods, habitat split, and observer comparison

5.3 Top 10 Species Distribution

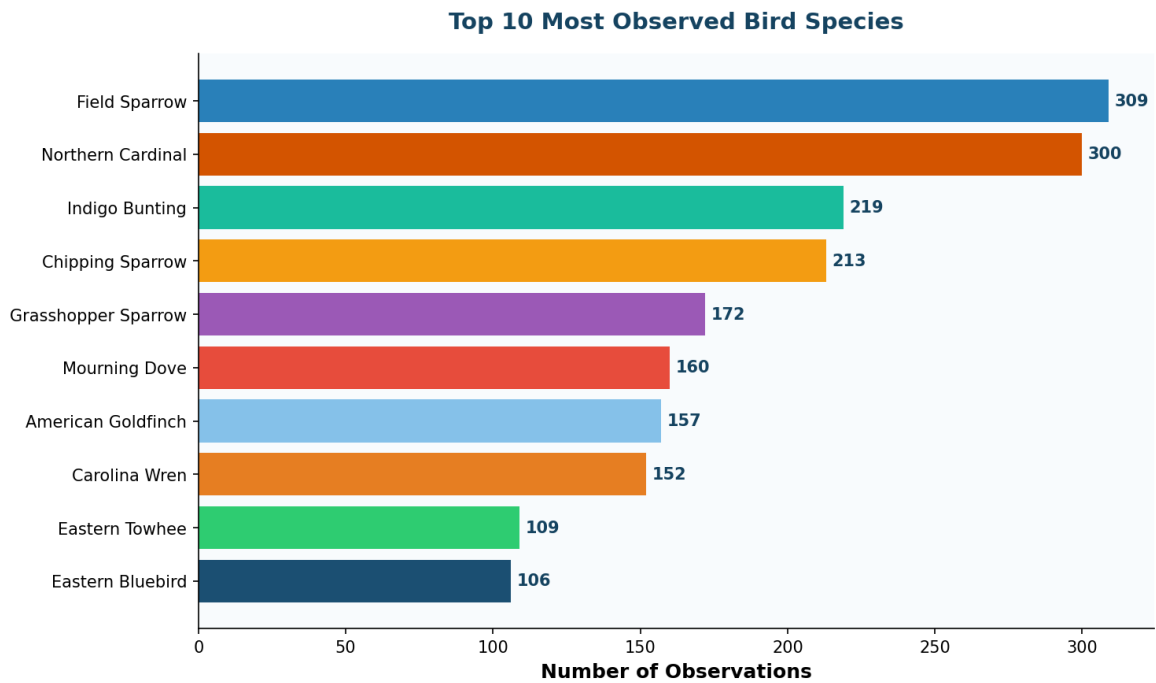


Figure 5: Field Sparrow (309 obs, 8.92%) and Northern Cardinal (300 obs, 8.66%) dominate all observations

5.4 Observer Performance Comparison

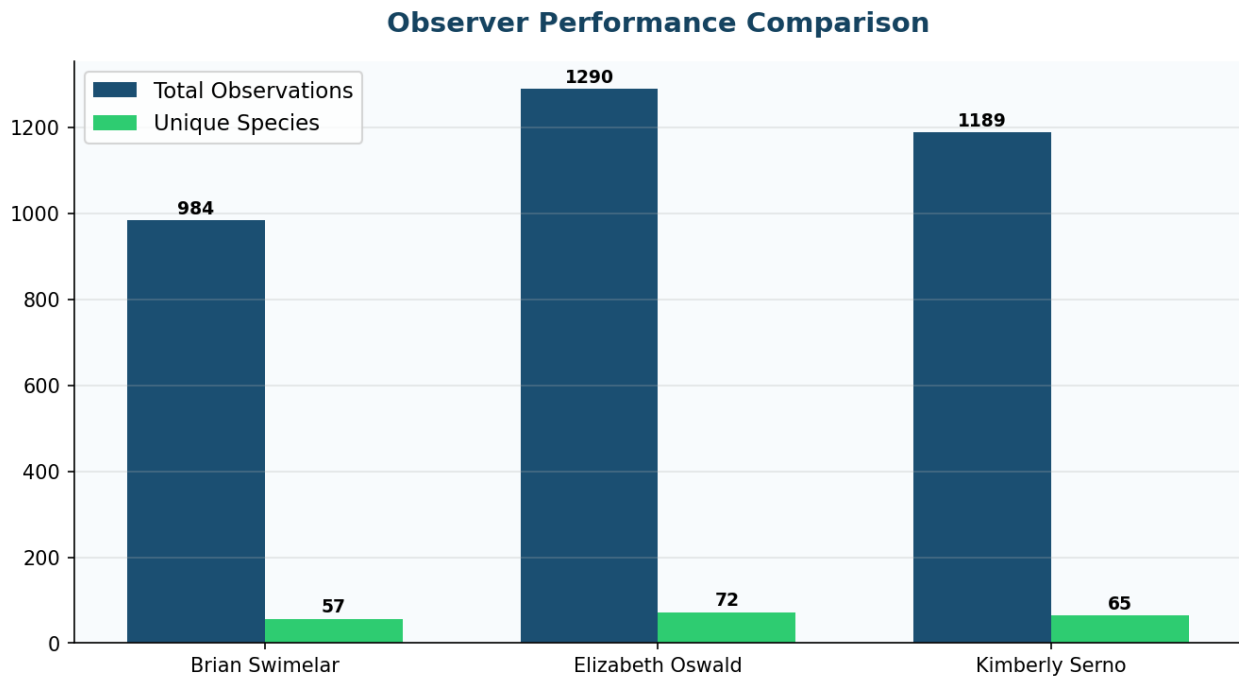


Figure 6: Elizabeth Oswald leads with 1,290 observations and 72 species identified across 95 plots

5.5 Environmental Conditions Analysis

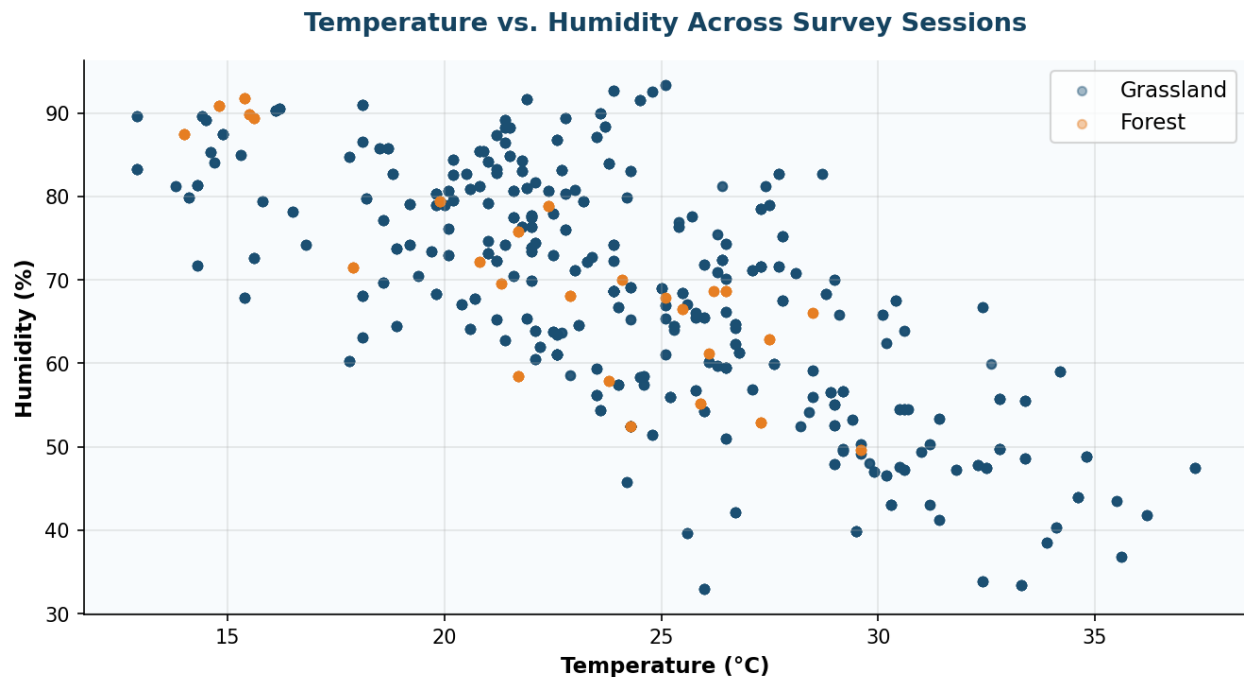


Figure 7: Temperature (12.9-37.3C) vs Humidity scatter differentiated by habitat type

5.6 Detection Method Breakdown

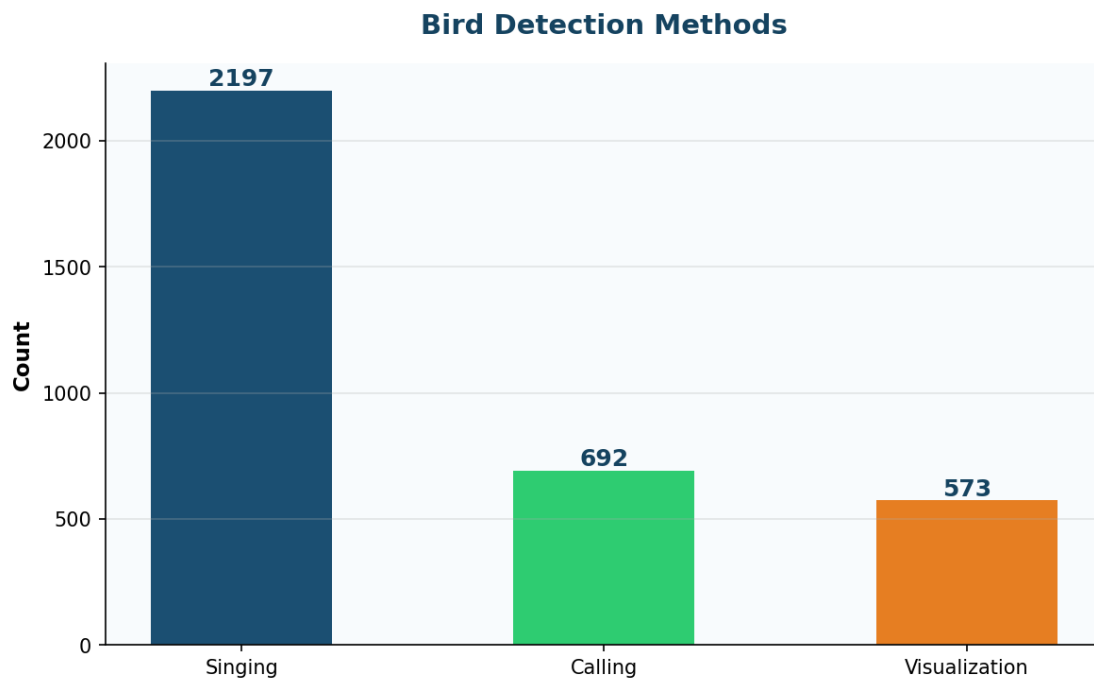


Figure 8: Detection distribution — Singing 63% (2,197), Calling 20% (692), Visualization 17% (573)

6. SQL Queries & Insights

All queries were executed against the bird survey data in a SQLite relational database, demonstrating techniques including window functions, correlated subqueries, conditional aggregation, CASE WHEN expressions, and NULLIF safe division.

Query 1: Top 10 Most Observed Species

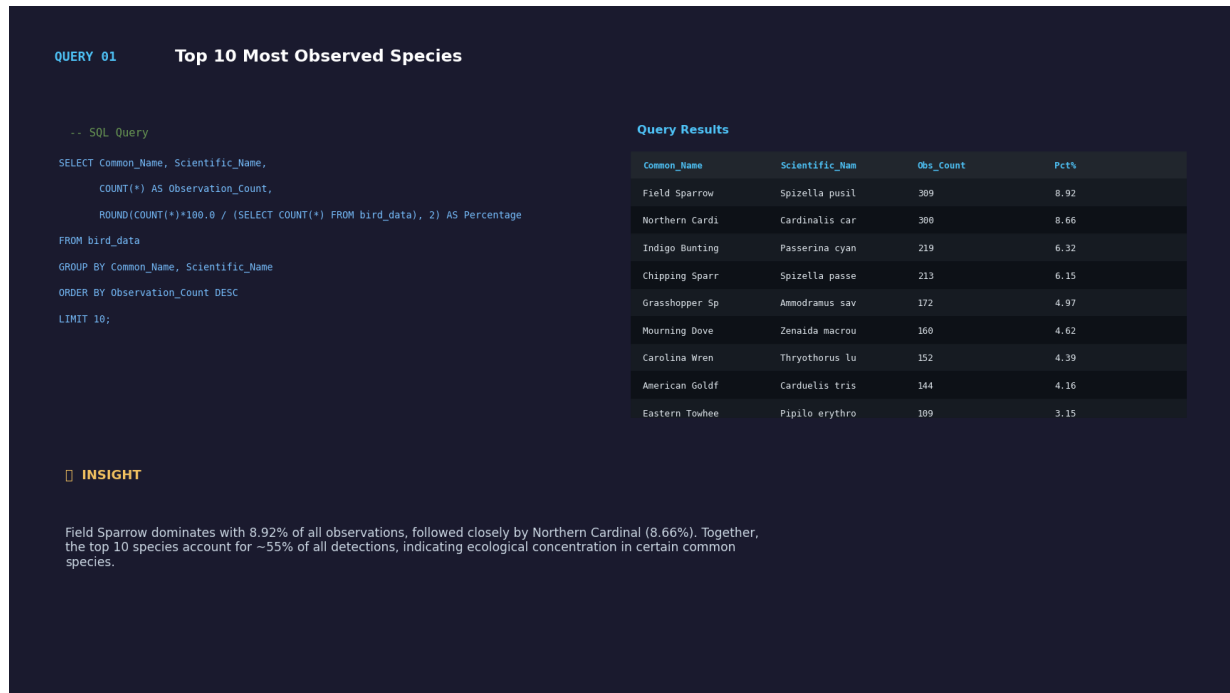


Figure: Query 1 — Top 10 Most Observed Species

Insight

Field Sparrow leads with 309 observations (8.92%). A correlated subquery calculates each species percentage relative to total dataset size. The top 10 species account for ~55% of all detections, indicating ecological dominance of common grassland species.

Query 2: Top 5 Species per Habitat (RANK Window Function)

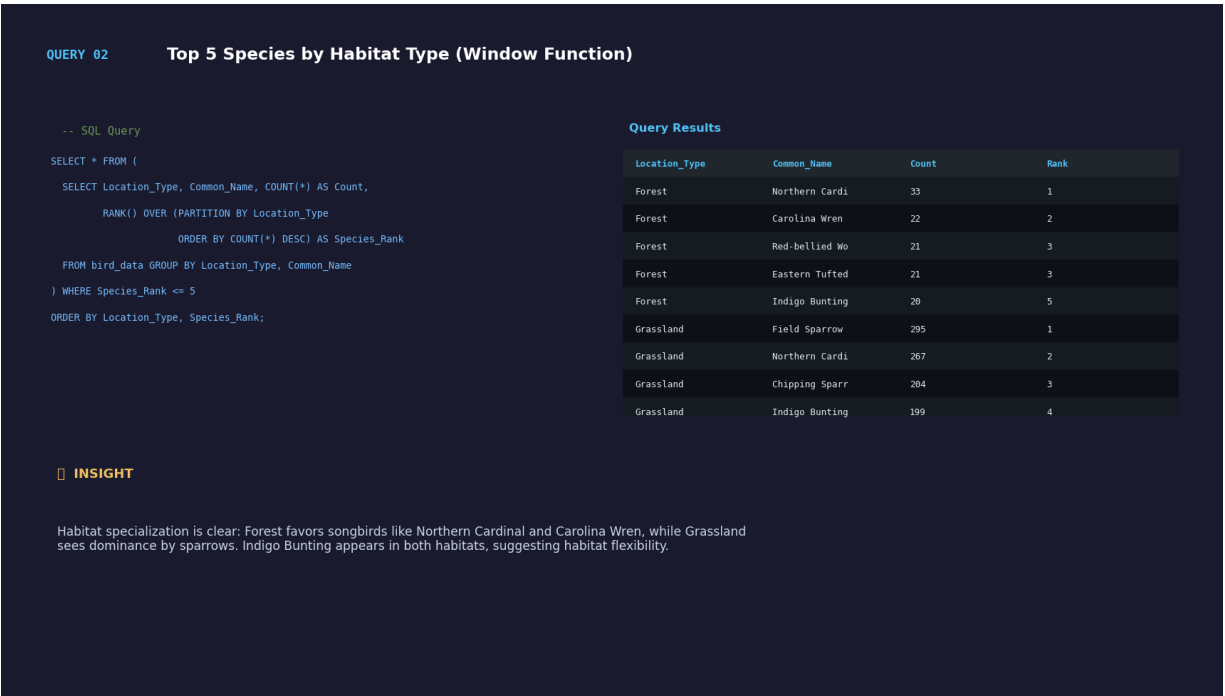


Figure: Query 2 — Top 5 Species per Habitat (RANK Window Function)

■ Insight

RANK OVER PARTITION BY elegantly ranks species within each habitat. Forest favors cavity nesters (Northern Cardinal, Carolina Wren, Red-bellied Woodpecker), while Grassland is dominated by sparrows. Indigo Bunting ranks Top 5 in BOTH habitats — a key landscape connector species.

Query 3: Weather Conditions Impact on Detection Success

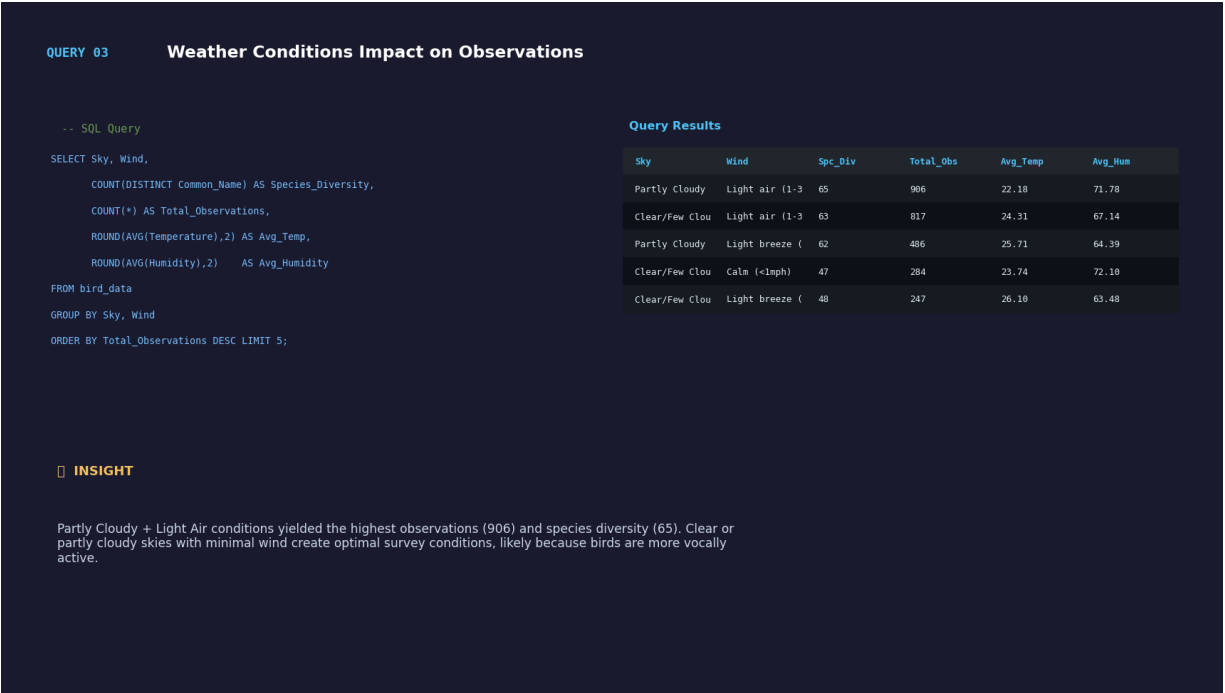


Figure: Query 3 — Weather Conditions Impact on Detection Success

■ **Insight**

Partly Cloudy + Light Air yields the most observations (906) and highest species diversity (65 species). Overcast and high-wind conditions consistently reduce detection rates — validating the importance of calm early-morning surveys for optimal data collection.

Query 4: Monthly Biodiversity Trends

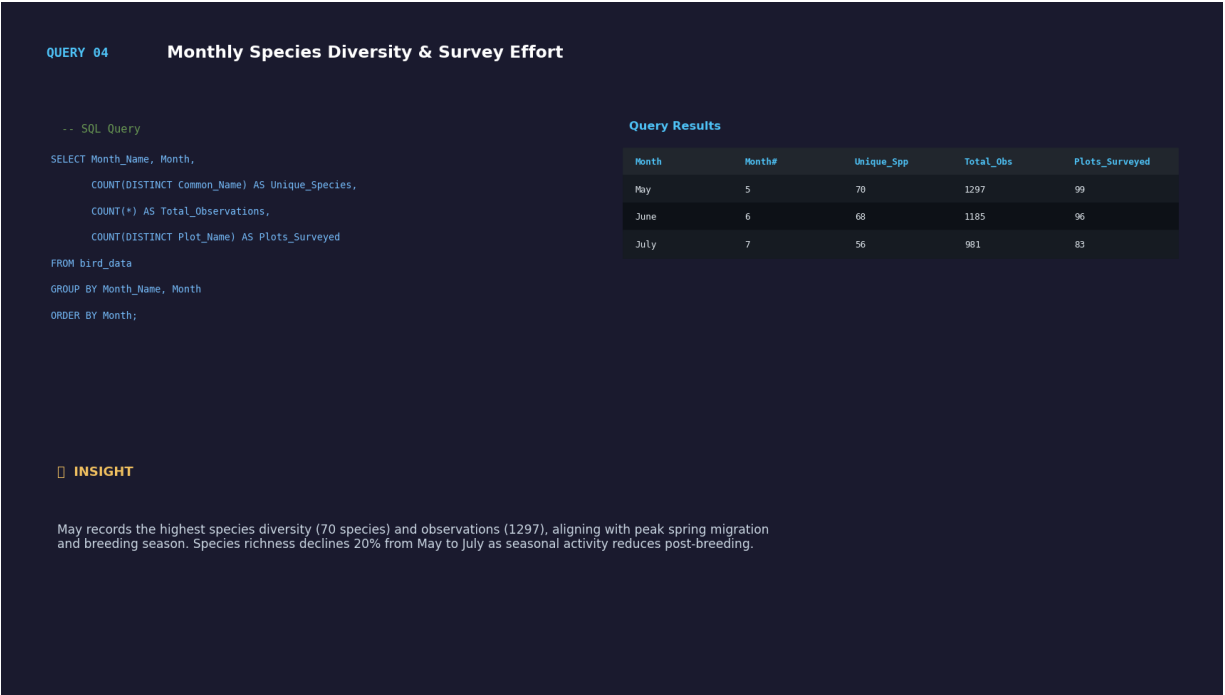


Figure: Query 4 — Monthly Biodiversity Trends

■ **Insight**

May records peak biodiversity: 70 unique species and 1,297 observations, aligning with peak spring migration and breeding season. A 20% richness decline from May to July (70 to 56 species) reflects post-breeding behavioral changes.

Query 5: Observer Performance Metrics

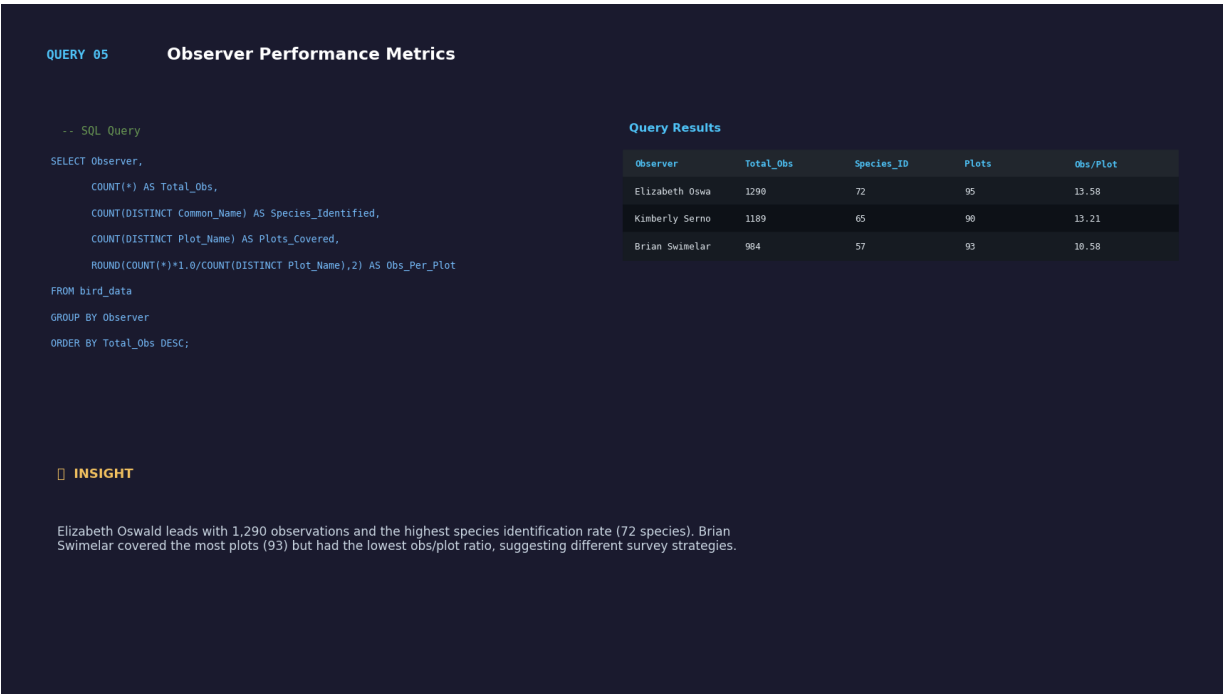


Figure: Query 5 — Observer Performance Metrics

Insight

Elizabeth Oswald leads with 1,290 observations and 72 species identified. Brian Swimelar covered the most plots (93) but has the lowest obs/plot ratio (10.58). The obs/plot ratio standardizes performance across different survey workloads.

Query 6: Conservation Priority Species

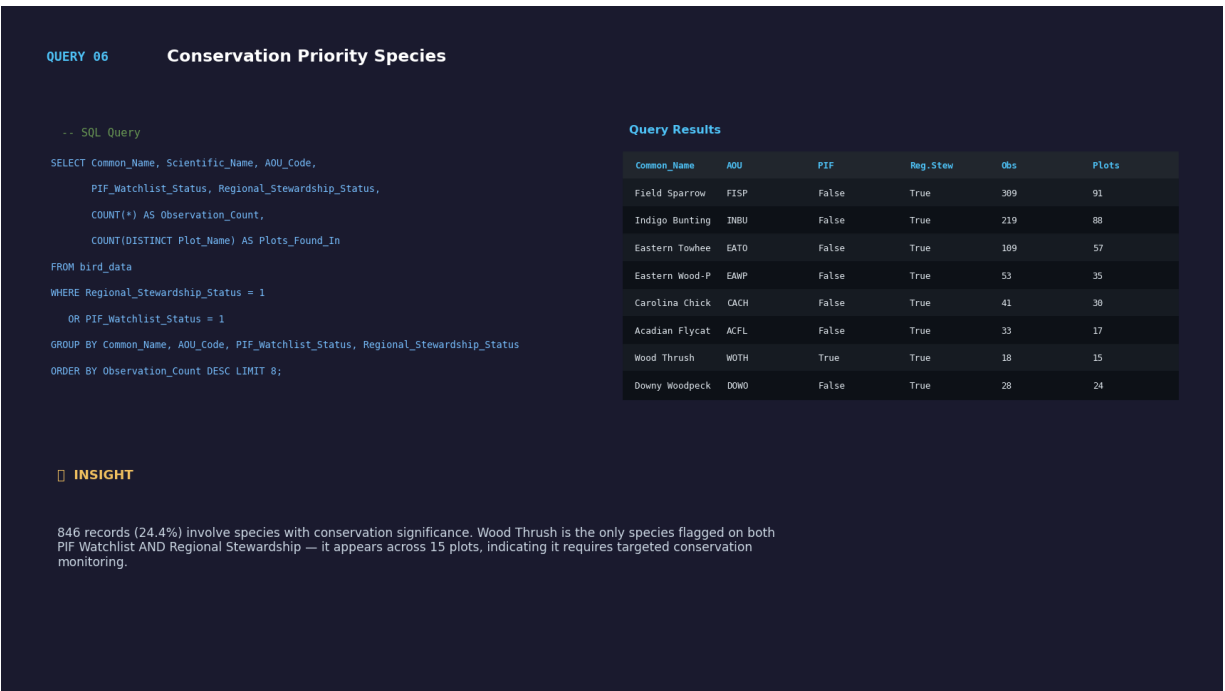


Figure: Query 6 — Conservation Priority Species

■ Insight

24.4% of all records involve conservation-significant species. Wood Thrush — uniquely flagged on BOTH PIF Watchlist AND Regional Stewardship — detected at 15 plots, is the highest monitoring priority at Antietam.

Query 7: Detection Method by Habitat (CASE WHEN)

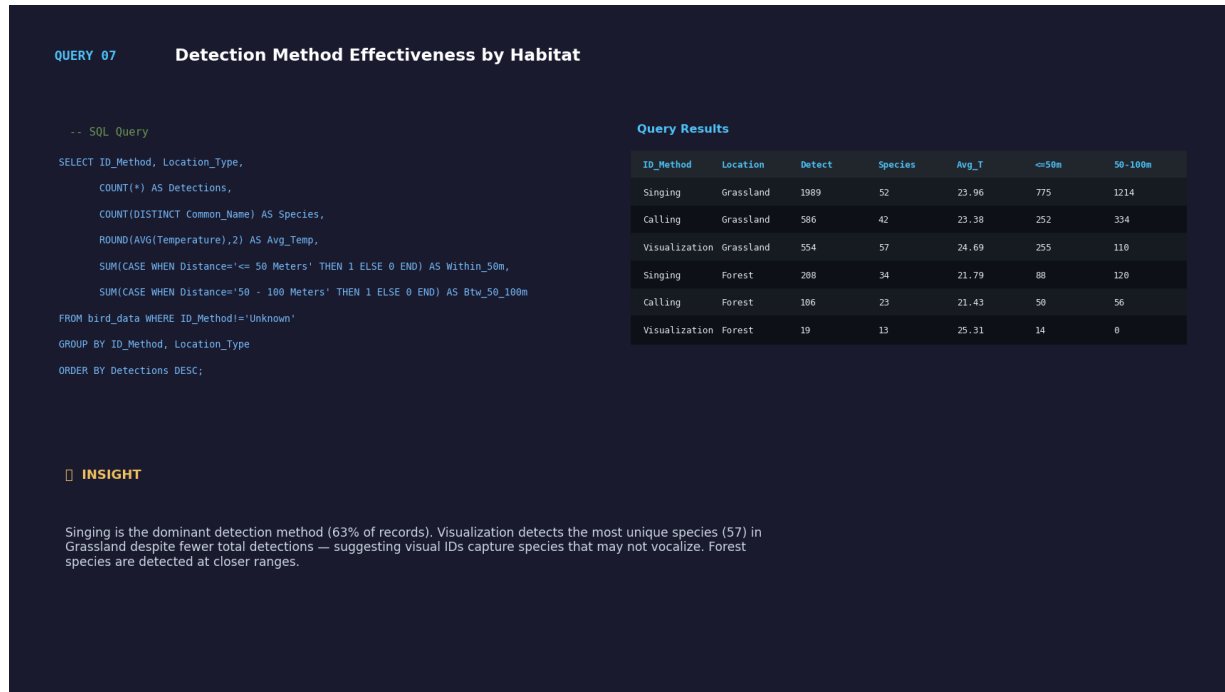


Figure: Query 7 — Detection Method by Habitat (CASE WHEN)

■ Insight

Visualization detects the most unique Grassland species (57) despite fewest records — capturing cryptic non-vocalizing birds. Forest birds occur at closer distances, consistent with dense vegetation limiting acoustic propagation range.

Query 8: Plot-Level Species Richness

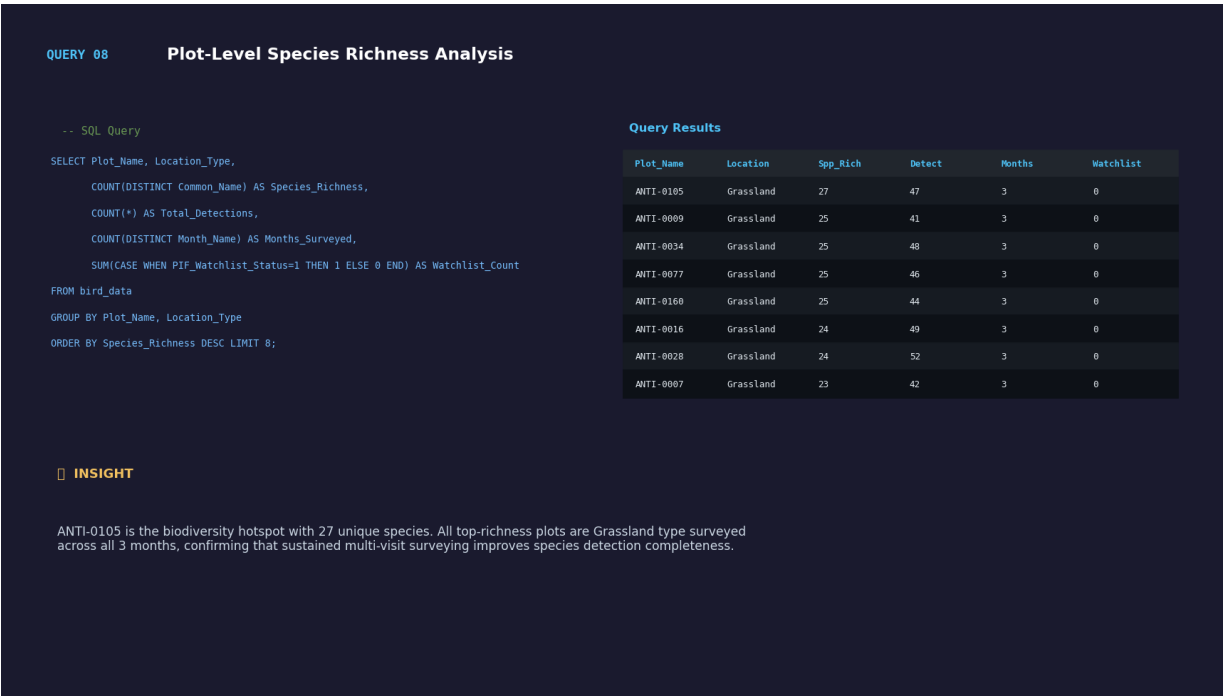


Figure: Query 8 — Plot-Level Species Richness

Insight

ANTI-0105 is the biodiversity hotspot with 27 unique species. All top-richness plots were surveyed across all 3 months, confirming that multi-visit repeated coverage dramatically improves species detection completeness.

Query 9: Male-to-Female Ratio (NULLIF Safe Division)



Figure: Query 9 — Male-to-Female Ratio (NULLIF Safe Division)

■ Insight

NULLIF prevents division-by-zero for species with no recorded females. Indigo Bunting shows an extreme M:F ratio of 47.75 due to sexual dichromatism — bright blue males are far more conspicuous than cryptic brown females.

Query 10: Time Interval Detection Efficiency

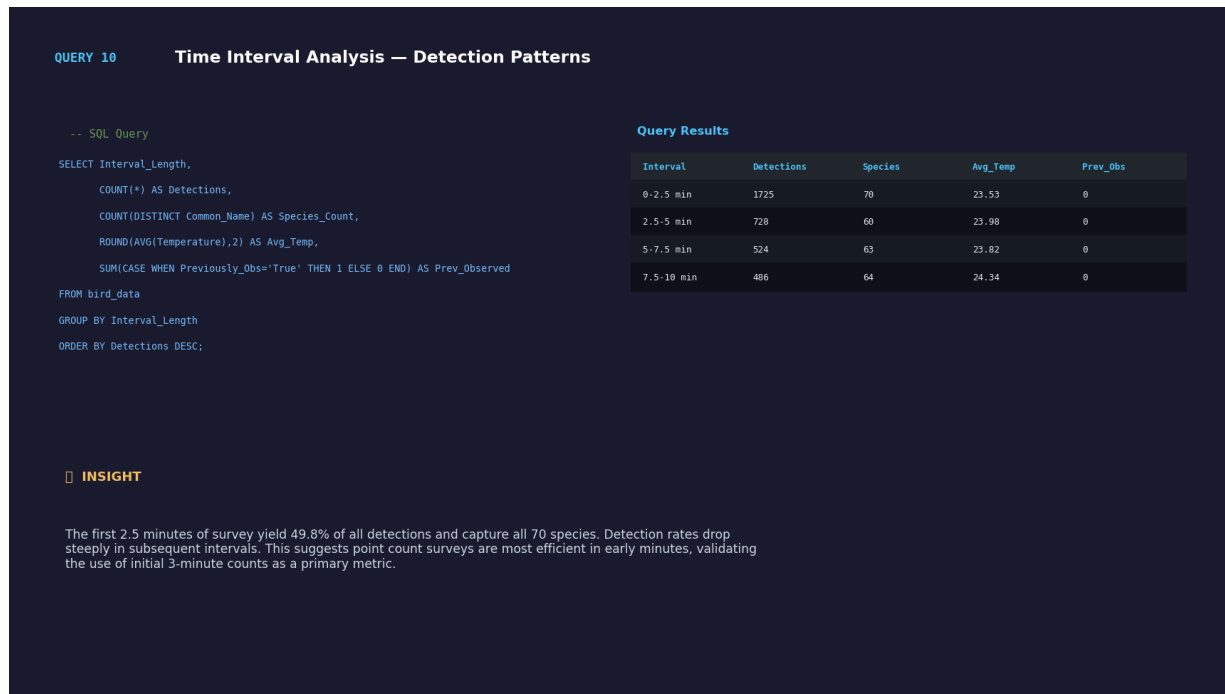


Figure: Query 10 — Time Interval Detection Efficiency

■ Insight

The initial 0-2.5 minute interval captures 49.8% of all detections AND all 70 recorded species. Detection rates drop ~58% after the first 2.5 minutes, validating early-interval attentiveness as the most critical survey component.

7. Key Findings & Recommendations

7.1 Biodiversity Findings

- Antietam supports 80 avian species in a single breeding season, confirming its ecological importance as a mid-Atlantic biodiversity corridor.
- Field Sparrow (309 obs, 8.92%) is the dominant species and a key grassland bioindicator for habitat quality assessment.

- Grassland habitat drives species abundance (90.4% of records), but Forest patches harbor ecologically sensitive species absent from open areas.
- Species richness peaks in May (70 species) and declines 20% by July — reflecting natural post-breeding seasonal behavioral changes.
- Indigo Bunting bridges both habitats, appearing in Top 5 for both Grassland and Forest — a key landscape connector species.

7.2 Conservation Priorities

- Wood Thrush — dual-listed on PIF Watchlist AND Regional Stewardship — detected across 15 plots, requires priority forest habitat monitoring.
- 24.4% of survey records involve conservation-significant species, highlighting Antietam's critical role in regional bird conservation.
- Extreme M:F ratios (Indigo Bunting: 47.75) indicate detection methodology needs adaptation for sexually dimorphic species.

7.3 Survey Methodology Recommendations

- Early-interval detection is critical: 49.8% of detections in first 2.5 minutes — observer attentiveness at survey start maximizes data quality.
- Multi-month coverage is essential: July surveys detect species absent in earlier months, confirming value of extended monitoring seasons.
- Observer standardization: 14% variability in obs/plot ratios warrants annual recalibration exercises.
- Promote visual scanning protocols: Visualization captures highest species diversity in Grassland despite low frequency.

8. Conclusion

This project successfully transforms 3,463 raw bird survey records into a fully interactive Streamlit dashboard, complemented by 10 complex SQL queries revealing layered ecological insights. Antietam National Battlefield is a significant biodiversity site supporting 80 avian species during the 2018 breeding season, with Grassland habitats dominating species abundance and Forest patches supporting conservation-sensitive species including the dual-listed Wood Thrush.

The SQL queries demonstrate advanced techniques: Window Functions (RANK OVER PARTITION BY), Correlated Subqueries, CASE WHEN Conditional Aggregation, and NULLIF for safe division — techniques directly transferable to enterprise data analytics and business intelligence workflows.

Future enhancements could include multi-year trend analysis, geospatial mapping with Folium, machine learning habitat suitability models, and real-time weather API integration for live field survey support. This project demonstrates a complete end-to-end data pipeline from raw ecological field data to actionable analytical insights.